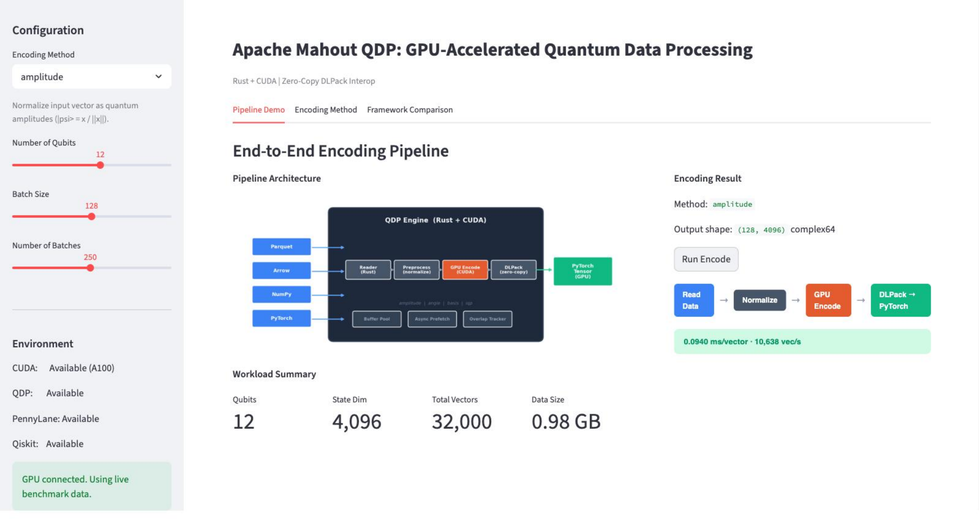
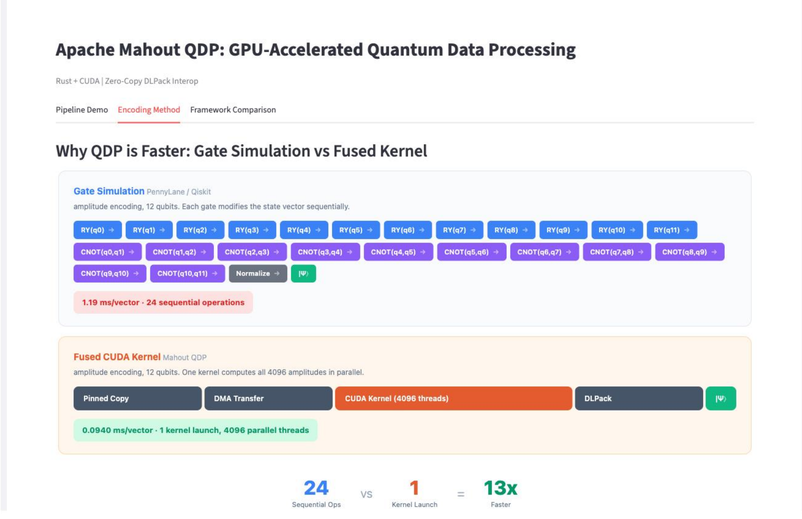
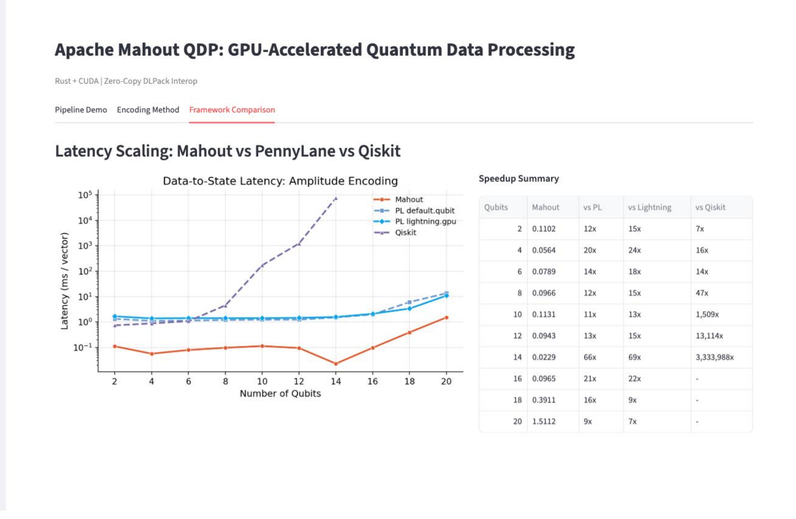


(a) Pipeline Demo



(b) Encoding Method



Qubits	Mahout	vs PL	vs Lightning	vs Qiskit
2	0.1102	12x	15x	7x
4	0.0564	20x	24x	16x
6	0.0789	14x	18x	14x
8	0.0966	12x	15x	47x
10	0.1131	11x	13x	1,509x
12	0.0943	13x	15x	13,114x
14	0.0229	66x	69x	3,333,988x
16	0.0965	21x	22x	-
18	0.3911	16x	9x	-
20	1.5112	9x	7x	-

(c) Framework Comparison